Fourth Problem Sheet
on
Complex Analysis
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Topic: Complex Differentiability

"Complex analysis is the good twin and real analysis the evil one: beautiful formulas and elegant theorems seem to blossom spontaneously in the complex domain."
-Peter Henrici

"Real analysis is the battlefield; complex analysis is the paradise."
-Paul Halmos

"The miracle of complex differentiability is that from one simple condition — the Cauchy-Riemann equations — an entire universe of rigidity and beauty emerges."
-Lars Ahlfors

"In the complex domain, differentiability once implies differentiability infinitely many times. Nature is more generous than we deserve."
-Henri Cartan

"The requirement of complex differentiability is so rigid, so demanding, that functions satisfying it reveal the deepest secrets of mathematics."
-Carl Ludwig Siegel

"Riemann's concept of the complex derivative unified geometry, analysis and physics in one stroke, an act of mathematical genius that may never be surpassed."
-Felix Klein

Problem 1

Let $q : \mathbb{C} \to \mathbb{C}$ and let $w \in \mathbb{C}$. Show that $\lim_{z \to w} q(z) = L$ if and only if $\lim_{z \to 0} q(z + w) = L$. Deduce that for any point z_0 in the domain of a function f, f is differentiable at z_0 if and only if

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists. In this case $f'(z_0)$ also equals the latter limit.

Problem 2

Show that $f(z)$ is holomorphic if and only if $(f(z^*))^*$ is holomorphic. Discuss whether the derivative of the conjugate of a function is the same as the conjugate of the derivative.

Problem 3

Let f and g be two functions such that $f(z_0) = g(z_0) = 0$, both differentiable at z_0 with $g'(z_0) \neq 0$. Prove that

$$\lim_{z \to z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

The latter is known as the **L'Hospital's rule** for complex functions.

Problem 4

We proved in class that $f(z) = z^n$ for $n = 0, 1, 2, \dots$ is an entire function. Use the Stolz-Fréchet criterion to prove that f is entire for each n natural by finding the function $\mathcal{E}(z)$ such that

$$f(z) = f(z_0) + A(z - z_0) + \mathcal{E}(z)(z - z_0)$$

with $\mathcal{E}(z) \to 0$ as $z \to z_0$. Does the complex number A coincides with the expected derivative of f? Use the same criterion to show that the same result holds for negative integers n at each point of $\mathbb{C} \setminus \{0\}$.

Problem 5

We say that a function f is **antiholomorphic** if it is the complex conjugate of a holomorphic function.

i) Show that an antiholomorphic functions satisfies the **Anti-Cauchy-Riemann equations** (also known as the **conjugate Cauchy-Riemann equations**)

$$\partial_x u = -\partial_y v \quad \text{and} \quad \partial_y u = \partial_x v.$$

ii) Is the converse also true?

iii) Can a complex function be holomorphic and anitholomorphic at the same time? If so, determine which type of functions, and if not, prove it.

Problem 6

Let $\mathcal{V} = C^2(U)$ with $U \subseteq \mathbb{R}$ be the linear space of functions $f : U \to \mathbb{R}$ at least class C^2. We define the **Laplace operator** $\Delta : C^2(U) \to C(U)$ as

$$\Delta \doteq \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

We say that a function $f \in \mathcal{V}$ is **harmonic** if $f \in \ker \Delta$.

Show that the Cauchy-Riemann operators factorise the Laplace operator via

$$\Delta = 4 \cdot \partial_z \partial_{z^*},$$

and thus deduce that any holomorphic function is harmonic. Conclude that the sum of a holomorphic function with an anitholomorphic function yields a harmonic function. Finally, write $z = x + ivt$ and hence deduce that the general solution to the one-dimensional wave equation

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 u}{\partial t^2}$$

is given by

$$u(x, t) = f(x + vt) + g(x - vt),$$

with f, g any two differentiable functions of their arguments.

Problem 7

By considering the gradients of the real and imaginary parts of a holomorphic function, show that the contour lines of the former and the latter intersect orthogonally. This result is represented graphically in Figure 1.

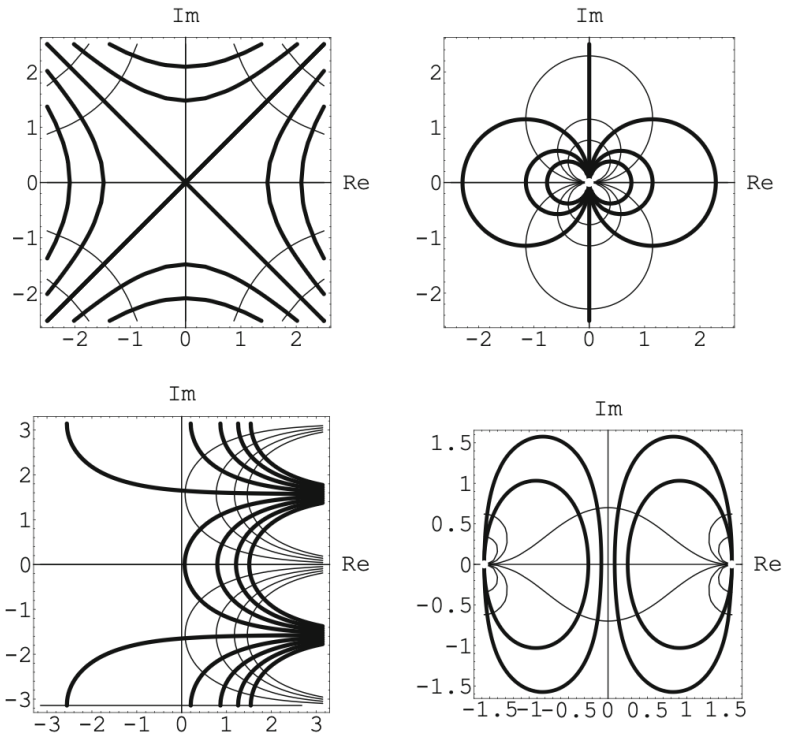


Figure 1: Orthogonal intersections of the real (thick trace) and imaginary (thin trace) parts of holomorphic functions z^2 (top left), $\frac{1}{z}$ (top right), e^z (bottom left), $\tan z$ (bottom right)

Problem 8

Let $z = x + iy$ and write x, y as functions of $z, \bar{z}$. Then deduce the Cauchy-Riemann operators by using the chain rule and treating $z, \bar{z}$ as independent variables. Verify that the Cauchy-Riemann operator $\partial_{\bar{z}}$ "differentiates with respect to $\bar{z}$ only" by showing that

i) $\partial_{\bar{z}} \bar{z}^n = n\bar{z}^{n-1}$ for $n \in \mathbb{N}$;
ii) $\partial_{\bar{z}}(\sin \bar{z} \cos z) = \cos \bar{z} \cos z$;
iii) $\partial_{\bar{z}}(z^2 \bar{z}) = z^2$;
iv) $\partial_{\bar{z}} e^{z\bar{z}} = z e^{z\bar{z}}$.

Problem 9

Show that the function

$$f(z) = \begin{cases} z^5/|z|^4 & z \neq 0, \\ 0 & z = 0, \end{cases}$$

satisfies the Cauchy-Riemann equations at $z = 0$, but at the same point it is not differentiable.